

Caleb O'Neal

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EDUCATION

University of Virginia

Bachelor of Science in Computer Engineering

Charlottesville, VA

Expected May 2028

Northern Virginia Community College

Associate of Science in Computer Science

Annandale, VA

June 2025

WORK EXPERIENCE

Intern

August 2024 – May 2025

Virginia Tech Qualcomm Thinkabit Lab

Falls Church, VA

- Programmed a 3-phase brushless DC motor simulation with a web interface to educate robotics students
- Assisted in lab tasks like cleaning, organizing, leading school groups, and moving the lab between buildings

Teaching Assistant

January 2025 – May 2025

Arlington Career Center

Arlington, VA

- Taught a college level statistics class as the primary instructor, covering the principles of statistical analysis
- Graded and reviewed students' assignments, including a poster session of students' individual research of a statistical relationship in real world datasets

High School Intern

June 2024 – August 2024

Department of Defense

Virginia

- Redesigned a department website to guide agency management to enterprise architecture visualization tools
- Migrated the team's Git repositories between Git hosts, ensuring commit history was preserved
- Collaborated with the team's Scrum Master to follow Agile software development best practices

PROJECTS

PyCon US 2024 Poster Presentation, NASA App Development Challenge

2024

- Developed a 3D interactive simulation of lunar surface exploration for the planned 2027 NASA Artemis III mission to the lunar south pole
- Presented our simulation to software industry professionals at the PyCon 2024 poster presentation session
- Collaborated with a team to develop the app and manage team finances and logistics

Martian Crater Data Analysis

- Analyzed a dataset of 350,000 Martian craters to find a statistical relationship between crater eccentricity and its latitude on Mars, likely as a result of volcanic activity on the Martian surface
- Programmed in SAS to perform statistical analysis and generate diagrams showing relevant data
- Presented results to peers for review and at a local community college for a larger audience

Captain, Programming Lead, Robotics Team

August 2022 – May 2025

- Managed a team of 50 with an annual budget of \$50,000 to design, build, wire, program, and test a robot designed for the 2024 and 2025 FIRST Robotics Competitions
- Redesigned the team management structure following significant growth in members
- Led a team of 10 programmers using GitHub's pull request and issue systems for code review

SKILLS

- **Programming:** Python, C, C++, Java, JavaScript, SAS, R, Rust, Julia, Zig
- **Tools:** Git, CI/CD, Linux, Docker
- **CAD:** Fusion 360, OnShape, Altium